

Estimation of Cerebral Blood Flow and Arterial Transit Time From Multi-Delay Arterial Spin Labeling MRI Using a Simulation-Based Supervised Deep Neural Network

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Background: An inherently poor signal-to-noise ratio (SNR) causes inaccuracy and less precision in cerebral blood flow (CBF) and arterial transit time (ATT) when using arterial spin labeling (ASL). Deep neural network (DNN)-based parameter estimation can solve these problems.

Purpose: To reduce the effects of Rician noise on ASL parameter estimation and compute unbiased CBF and ATT using simulation-based supervised DNNs.

Study Type: Retrospective.

Population: One million simulation test data points, 17 healthy volunteers (five women and 12 men, 33.2 ± 14.6 years of age), and one patient with moyamoya disease.

Field Strength/Sequence: 3.0 T/Hadamard-encoded pseudo-continuous ASL with a three-dimensional fast spin-echo stack of spirals.

Assessment: Performances of DNN and conventional methods were compared. For test data, the normalized mean absolute error (NMAE) and normalized root mean squared error (NRMSE) between the ground truth and predicted values were evaluated. For in vivo data, baseline CBF and ATT and their relative changes with respect to SNR using artificial noise-added images were assessed.

Statistical Tests: One-way analysis of variance with post-hoc Tukey's multiple comparison test, paired t-test, and the Bland-Altman graphical analysis. Statistical significance was defined as $P < 0.05$.

Results: For both CBF and ATT, NMAE and NRMSE were lower with DNN than with the conventional method. The baseline values were significantly smaller with DNN than with the conventional method (CBF in gray matter, 66 ± 10 vs. 71 ± 12 mL/100 g/min; white matter, 45 ± 6 vs. 46 ± 7 mL/100 g/min; ATT in gray matter, 1424 ± 201 vs. 1471 ± 154 msec). CBF and ATT increased with decreasing SNR; however, their change rates were smaller with DNN than were those with the conventional method. Higher CBF in the prolonged ATT region and clearer contrast in ATT were identified by DNN in a clinical case.

Data Conclusion: DNN outperformed the conventional method in terms of accuracy, precision, and noise immunity.

Evidence Level: 3

Technical Efficacy: Stage 1

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Arterial spin labeling (ASL) is a noninvasive perfusion imaging technique that uses magnetically inverted spins in arterial blood as endogenous tracers.¹ Perfusion images are commonly computed by subtracting two images in opposite magnetization regimes: label and control.² Multi-delay ASL can quantify cerebral blood flow (CBF) and arterial transit time (ATT), which is the time required for labeled spins to traverse the spatial gap between the labeling plane and the brain tissue, but also provides functional information on cerebral hemodynamics.³ CBF and ATT are clinically used to diagnose various diseases.^{4–7}

Although there are many advantages to the clinical use of ASL, its intrinsically low signal-to-noise ratio (SNR) degrades the accuracy and the precision of CBF and ATT.^{8,9} To solve this problem, researchers have endeavored to improve and optimize the bolus design of a multi-delay ASL.^{9–11} Dai et al. developed a robust ATT estimation strategy, the signal weighted-delay (WD) method, which avoids using nonlinear least-squares (LS) fitting.¹² Many ASL studies have employed this method.^{3,7,9,13–15}

As ASL-MRI has a low temporal resolution, it is difficult to acquire abundant temporal information with sufficient SNR in a clinical routine framework, even with a time-efficient technique.¹⁶ Insufficient temporal resolution makes the parameter estimation more vulnerable. These disadvantages of ASL hamper the clinical use of multi-delay ASL routinely, even though it provides valuable hemodynamic information.

Deep learning (DL), a subtype of machine learning, has drastically improved performance in classification and image recognition.^{17,18} A neural network is a basic DL algorithm. It has been reported that the accuracy and the precision of intravoxel incoherent motion (IVIM) fitting are improved by a deep neural network (DNN).^{19–22} Like ASL, IVIM fitting suffers from intrinsically poor SNR of diffusion-weighted images.²² Thus, DNNs may be able to overcome problems originating from the poor SNR of the ASL. However, there are only a few reports on the application of DL to the technical aspects of ASL, albeit a promising tool.^{23–27} Previous studies addressed the technical issues of ASL, such as the spatial resolution, SNR, and partial volume effects. No study applied DNNs to ASL parameter estimation, although they obtained remarkable results in IVIM parameter estimation.^{19–22} Robust and accurate reconstruction techniques may facilitate clinical use of multi-delay ASL. The present study aimed to reduce the effects of Rician noise on ASL parameter estimation and compute unbiased CBF and ATT from multi-delay ASL data using simulation-based supervised DNNs.

Materials and Methods

This study was approved by the institutional review board, and informed consent was obtained from all the participants. Figure 1 illustrates the flow of this study. First, the simulation data were calculated. Second, the best-performing architecture was selected from

among the candidates (Fig. 2). Subsequently, using the simulation data, the performance was compared between the selected DNN and the conventional method that used the WD method and the LS solution for estimating the ATT and the CBF, respectively. Performance was also evaluated in healthy volunteers ($n = 17$) and a patient with moyamoya disease.

All images were acquired using a 3.0 T MRI (Discovery MR 750; GE Healthcare, Waukesha, WI) with a 32-channel head coil. The generation of simulation data, implementation of DNNs, and image reconstruction were conducted using MATLAB 2022a (MathWorks, Natick, MA). All experiments were performed on a personal computer with an Intel(R) Core(TM) i9-10980XE CPU at 3.00 GHz and an NVIDIA GeForce RTX 3090.

In this study, the previously proposed method,⁹ composed of Hadamard 3 delay-encoding with an extra step and long-label long-delay (LLLD) acquisition, was adopted (hereafter referred to as the combined approach). Five boluses were collected, where the post-labeling delay (PLD) and the labeling duration (LD) of boluses 1 to 5 were as follows: (PLD [msec], LD [msec]) = (700, 1333), (2033, 1333), (3366, 1333), (700, 4000), and (3000, 4000).

Preparing Simulation Data

All networks were trained in a supervised manner using simulation data owing to the availability of the ground truths of the estimators. The ground truths of CBF and ATT were set in the range of 20–90 mL/100 g/min and 500–3000 msec, respectively, from which 10,000, 200, and 10,000 pairs of CBF and ATT were uniformly and randomly sampled for training, validation, and test datasets, respectively (Fig. 1a).^{8,9,28} For all pairs of CBF and ATT, the ASL signals were calculated using the single-compartment model:

$$\Delta M = 2\alpha\beta T_{1t} \frac{M_0}{\lambda} \cdot f \cdot \exp\left(\frac{-\delta}{T_{1a}}\right) \cdot \left[\exp\left(\frac{-\max(w-\delta, 0)}{T_{1t}}\right) - \exp\left(\frac{-\max(\tau+w-\delta, 0)}{T_{1t}}\right) \right], \quad (1)$$

where ΔM is the ASL signal, α is the labeling efficiency (0.85), β is the background suppression efficiency (0.75), M_0 is the equilibrium magnetization of brain tissue, λ is the blood partition coefficient (0.9), f is the CBF, T_{1t} is the T_1 of brain tissue (1.2 sec), T_{1a} is the T_1 of arterial blood (1.66 sec), δ is the ATT, w is the PLD, and τ is the LD.

The calculated ASL signals were scaled by a factor of 100000 (i.e., $\Delta M/M_0 = 1.0\%$ corresponds to a signal intensity of 1000). They were resolved into four components (Fig. 1b): a real part of the control (R_C), an imaginary part of the control (I_C), a real part of the label (R_L), and an imaginary part of the label (I_L). The noise was randomly generated from independent and identical Gaussian distributions and added to R_C , I_C , R_L , and I_L (Fig. 1c). The standard deviation (SD) of the Gaussian distribution was set so that the relative SNR (rSNR) ranged from 2 to 20, at which 500, 500, and 100 levels of SDs were defined with even spacing for the training,

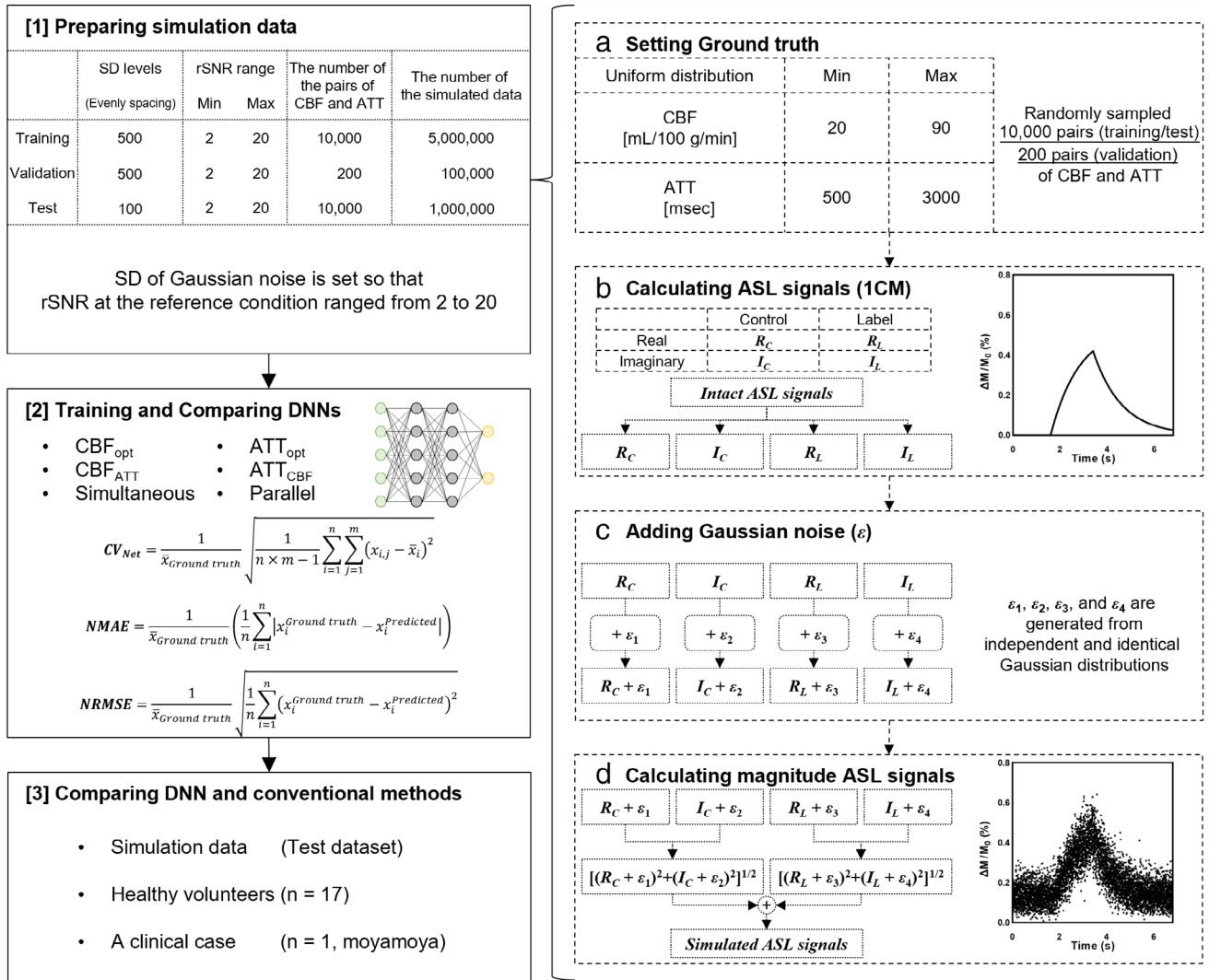


FIGURE 1: Flowchart of the current study. Subsections (a–d) illustrate how to generate noisy ASL signals. CBF_{opt}, ATT_{opt}, CBF_{ATT}, ATT_{CBF}, Simultaneous, and Parallel are the network names. Details of them are described in the text. SD, standard deviation; rSNR, relative signal-to-noise ratio; CV_{NET}, coefficient of variation of the network; NMAE, normalized mean absolute error; NRMSE, normalized root mean squared error; 1CM, the single-compartment model.

validation, and test datasets, respectively. Finer rSNR levels were set for the training dataset compared to those the test dataset to improve the prediction performance of DNNs. rSNR was defined based on the ASL signal at the reference condition (PLD = 2000 [msec], LD = 1800 [msec], CBF = 50 [mL/100 g/min], and ATT = 1600 [msec]).² To simulate the Rician noise distribution of magnitude ASL signals, magnitude control and label signals were calculated using the square root of the sum of squares of the real and the imaginary parts (Fig. 1d). Subtraction was reproduced as the sum of the control and the label signals. The total number of data points (product of the number of SD levels and the number of pairs of CBF and ATT) was five million, one hundred thousand, and one million for the training, validation, and test datasets, respectively.

Selection of the Best-Performing Network

Six architectures were devised, two each for estimating CBF (Fig. 2a, c), ATT (Fig. 2b,d), and CBF and ATT simultaneously (Fig. 2e,f). All the networks were composed of an input layer, fully connected hidden layers with an exponential linear unit activation function,²⁹

and an output layer. All the hidden layers had the same number of neurons. The inputs were $\Delta M/M_0$ after scaling up in the same way as mentioned above, and the outputs were the prediction parameters. Adam optimizer³⁰ with constant learning rate of 0.001, He initialization,³¹ and mean squared error loss were employed. Early stopping was used to terminate the training when the validation loss did not improve for 10 consecutive iterations.

The characteristics of DNNs were as follows: CBF_{opt}, the network dedicated to predicting CBF alone (Fig. 2a); ATT_{opt}, the network dedicated to predicting ATT alone (Fig. 2b); CBF_{ATT}, the CBF-dedicated network additionally using a pre-estimated ATT as an extra input (blue branch line in Fig. 2c); ATT_{CBF}, the ATT-dedicated network additionally using a pre-estimated CBF as an extra input (blue branch line in Fig. 2d); Simultaneous, CBF and ATT were predicted (Fig. 2e); Parallel, the CBF-dedicated and the ATT-dedicated networks were incorporated into a single network. Each parameter was predicted by the optimized subnetworks, but the loss function was concurrently assessed for CBF and ATT (Fig. 2f). CBF_{ATT} and ATT_{CBF} were introduced after selecting CBF_{opt} and ATT_{opt} (Supporting Information).

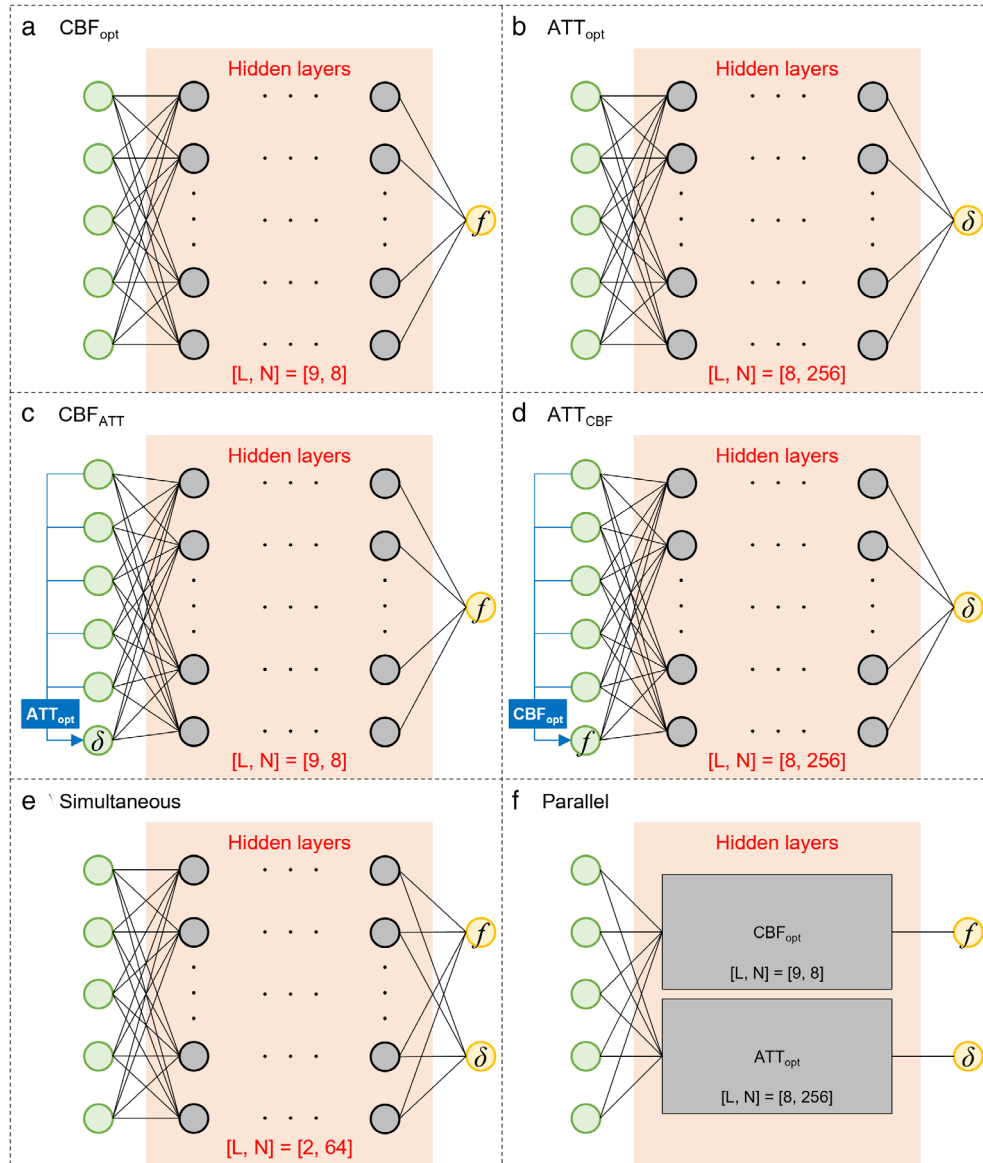


FIGURE 2: Schematic diagrams of deep neural network (DNN) architectures. All networks are composed of an input layer (green circles), fully connected hidden layers (gray circles), and an output layer (orange circles). The number of hidden layers and neurons (presented as $[L, N]$ in the figure) is preliminarily defined by grid search. Brief explanations of each DNN are as follows: the dedicated network to CBF estimation (CBF_{opt} , a), the dedicated network to ATT estimation (ATT_{opt} , b), the CBF-dedicated network using pre-estimated ATT (CBF_{ATT} , c), the ATT-dedicated network using pre-estimated CBF (ATT_{CBF} , d), simultaneous estimation of CBF and ATT (e), and the parallel network (f). CBF_{ATT} (c) and ATT_{CBF} (d) were introduced after selecting CBF_{opt} (a) and ATT_{opt} (b). Blue lines in (c) and (d) indicate the pathways for pre-estimation of ATT or CBF, respectively. Details of the DNNs are illustrated in the text.

A preliminary grid search was conducted to select the optimal architectural configuration, i.e., the number of hidden layers and neurons, of each DNN (Table S1 in the Supplemental Material). Similar to the preliminary grid search, the optimized DNNs were compared based on the normalized coefficient of variation over repeated training (CV_{Net}), normalized mean absolute error (NMAE), and normalized root mean squared error (NRMSE). DNNs (CBF_{opt} , CBF_{ATT} , *Simultaneous*, and *Parallel* for CBF and ATT_{opt} , ATT_{CBF} , *Simultaneous*, and *Parallel* for ATT) were compared based on the summed rankings of the three metrics, whereby each DNN received rankings of the respective metrics and subsequently summed.²¹ The summed ranks of 3 (i.e., $CV_{Net}/MAE/NRMSE = 1/1/1$) and

12 (i.e., $CV_{Net}/MAE/NRMSE = 4/4/4$) were the theoretical best and worst cases, respectively. The minimum summed ranking was adopted as the champion DNN.

To evaluate the consistency of estimators, each network was trained 20 times using an identical dataset with random initialization of every training. CV_{Net} was used as a metric for precision.²¹

$$CV_{Net} = \frac{1}{\bar{x}_{Ground\ truth}} \sqrt{\frac{1}{n \times m - 1} \sum_{i=1}^n \sum_{j=1}^m (x_{i,j} - \bar{x}_i)^2}, \quad (2)$$

where $\bar{x}_{Ground\ truth}$ is the mean of the ground truth (CBF or ATT), n is the number of data points (one million), m is the number of

repeated trainings,²⁰ $x_{i,j}$ is the j th repeated prediction of the i th data, $\bar{x}_i$ is the mean of the repeated predictions of the i th data.

The NMAE and NRMSE between the ground truth and the predicted values were used to evaluate accuracy.

$$\text{NMAE} = \frac{1}{\bar{x}_{\text{Ground truth}}} \left(\frac{1}{n} \sum_{i=1}^n |x_i^{\text{Ground truth}} - x_i^{\text{Predicted}}| \right), \quad (3)$$

$$\text{NRMSE} = \frac{1}{\bar{x}_{\text{Ground truth}}} \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i^{\text{Ground truth}} - x_i^{\text{Predicted}})^2}, \quad (4)$$

where $\bar{x}_{\text{Ground truth}}$ is the mean of the ground truth (CBF or ATT), n is the number of data points (one million), $x_i^{\text{Ground truth}}$ is the i th ground truth, $x_i^{\text{Predicted}}$ is the i th predicted datum.

Comparison of the DNN and the Conventional Method: A Simulation Study

The conventional method estimated ATT using WD,¹² where observed WD values were converted to ATT by numerical comparison to the theoretical WD values.

$$\text{WD}(\delta) = \frac{\left[\sum_{i=1}^n w_i \cdot \Delta M(\delta, w_i) \right]}{\left[\sum_{i=1}^n \Delta M(\delta, w_i) \right]}, \quad (5)$$

where ΔM is the ASL signal, δ is the ATT, w_i is the i th PLD, and n is the number of delays.

CBF was calculated from ASL signals and pre-estimated ATT using the LS solution⁹:

$$\hat{f} = (A^T A)^{-1} A^T B, \quad (6)$$

where $\hat{f}$ is the least squares solution (CBF) of the linear problem $B = Af$, A is a column vector of coefficients (C) of each bolus from Eq. 1 except for f ($A \triangleq [C_1 \cdots C_n]^T$), B is a column vector of ASL signals ($B \triangleq [\Delta M_1 \cdots \Delta M_n]^T$), and n is the number of delays.

Using the test dataset, the performances of the DNN (same architecture as CBF_{opt} in Fig. 2a and ATT_{opt} in Fig. 2b) and conventional methods were compared in the same way as in the previous section, except for CV_{Net} , because the conventional method was deterministic ($\text{CV}_{\text{Net}} = 0$). The CBF predicted by the DNN (CBF_{DNN}), the CBF calculated by the LS using the pre-estimated ATT by the DNN ($\text{CBF}_{\text{LS}} - \text{ATT}_{\text{DNN}}$), and the CBF calculated by the LS using the pre-estimated ATT by the WD ($\text{CBF}_{\text{LS}} - \text{ATT}_{\text{WD}}$) were compared. Moreover, the ATT predicted by the DNN (ATT_{DNN}) and the ATT calculated by the WD (ATT_{WD}) were compared.

Comparison of the DNN and the Conventional Method: A Healthy Volunteer Study

All images were the same as those collected in a previous study using a combined approach.⁹ The original SNRs of the volunteer data

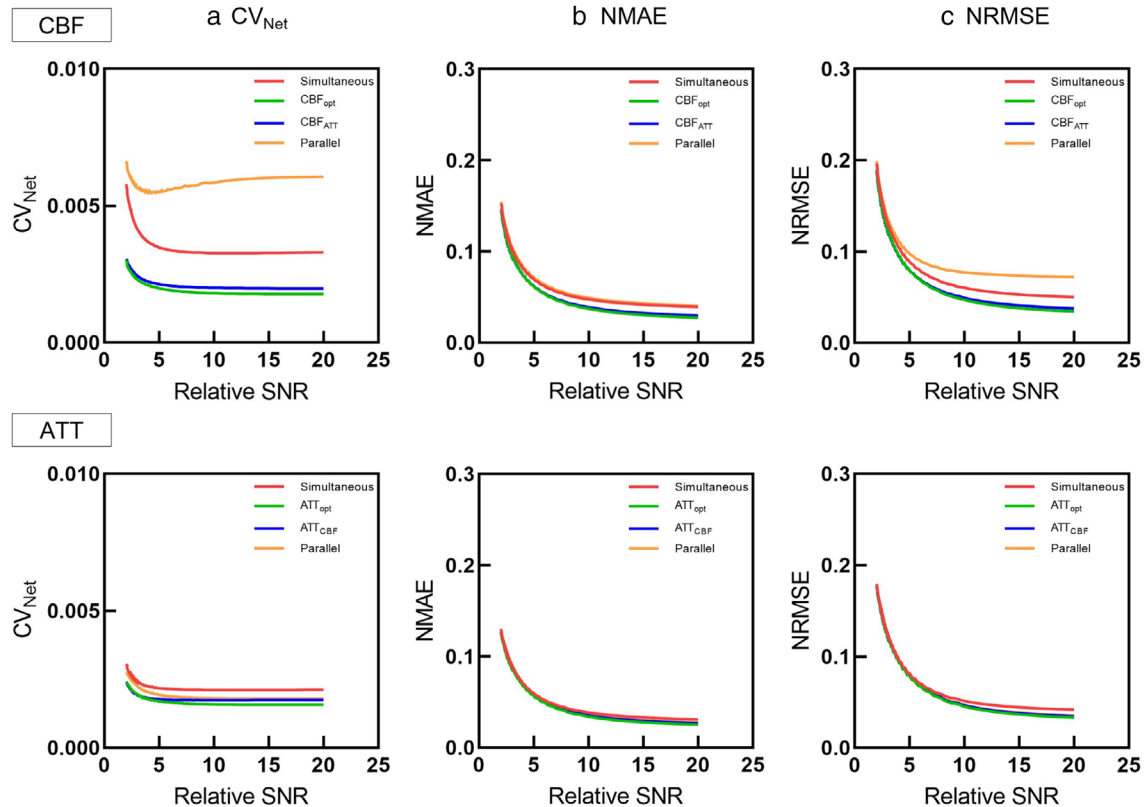


FIGURE 3: The behavior of the three metrics against the relative signal-to-noise ratio (SNR) evaluated using the test dataset. The upper and lower rows show the results of CBF and ATT, respectively. The left (a), center (b), and right (c) columns present the coefficient of variation of the network (CV_{Net}), the normalized mean absolute error (NMAE), and the normalized root mean squared error (NRMSE), respectively.

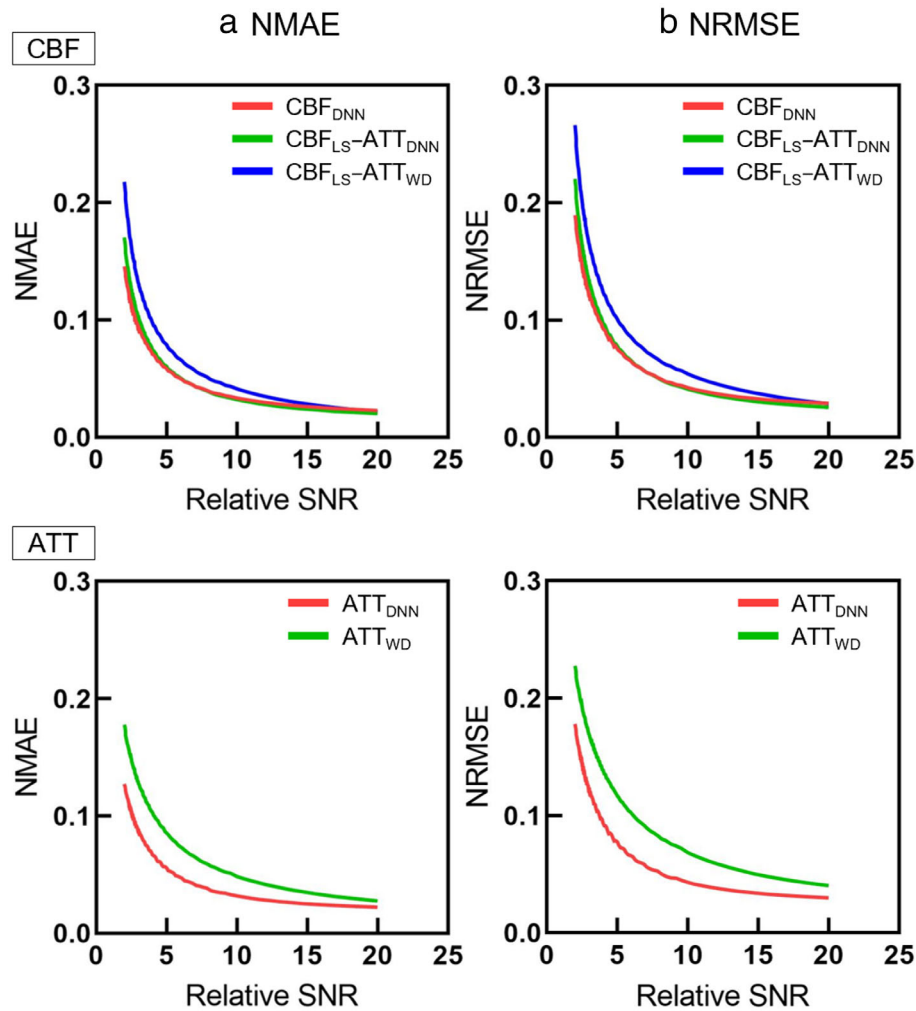


FIGURE 4: The results of the normalized mean absolute error (NMAE, a) and the normalized root mean squared error (NRMSE, b) of the deep neural network (DNN) and the conventional methods. The upper and lower rows show the results of CBF and ATT, respectively. Both metrics indicate that the estimation performance of the DNN is superior to that of the conventional method.

were determined to verify whether they were consistent with the simulation data, and the results are summarized in Supporting Information (Fig. S1 and Table S2 in the Supplemental Material). Similar to the simulation data, Rician noise with nine levels of SDs, which corresponded to an rSNR at the reference condition (PLD/LD/ATT = 2000/1800/1600 [msec] and CBF = 50 [mL/100 g/min]) of 2–10 (equally spaced), was artificially added to the observed ASL signals to synthesize noisy ASL images. CBF and ATT maps were computed from the original and synthesized images using the DNN and conventional methods. Differences in CBF and ATT maps between intact and noisy images at the respective rSNR levels were also calculated.

The imaging conditions of pseudo-continuous labeling³² with three-dimensional (3D) fast spin-echo stack of spirals were as follows: 6225 msec and 8525 msec repetition time (TR) for the Hadamard and the LLLD, respectively; 10.5 msec echo time (TE); 240 mm field-of-view (FOV); 4 mm thickness; 512 points with 6 interleaves; 1 average; no vascular suppression. 3D T₁-weighted images were also acquired (6.2 msec TR; 2.1 msec TE; 1 mm thickness; 240 × 216 mm FOV; 240 × 216 matrix; 15° flip angle; 1 average; 500 msec inversion time).

All ASL images were co-registered to 3D-T₁ images and spatially normalized to the canonical template using SPM12 (Wellcome

Center for Human Neuroimaging, UCL). Whole gray matter (WGM) and whole white matter (WWM) masks (probability threshold of 75%) were created from the 3D-T₁ images. CBF and ATT values were determined for WGM and WWM.

Noise immunity was assessed by relative changes in CBF and ATT against the SDs of artificially added noise, where CBF and ATT without added noise (i.e., zero SD) were considered as references. Baseline CBF and ATT calculated from images without artificial noise were compared. With intact images, the NRMSE between the measured signals and the signals calculated by substituting the estimated CBF and ATT for f and δ in Eq. 1 were computed for four pairs: (CBF_{DNN}, ATT_{DNN}), (CBF_{DNN}, ATT_{WD}), (CBF_{LS}, ATT_{DNN}), and (CBF_{LS}, ATT_{WD}), as residuum.¹⁹ The residuum was calculated as a metric for evaluating the validity of the baseline values, which were calculated from no added noise images, and they were determined in WGM and WWM.

Comparison of the DNN and the Conventional Method: A Clinical Case

The DNN and conventional methods were applied to a patient with moyamoya disease (68-year-old man) to demonstrate clinical

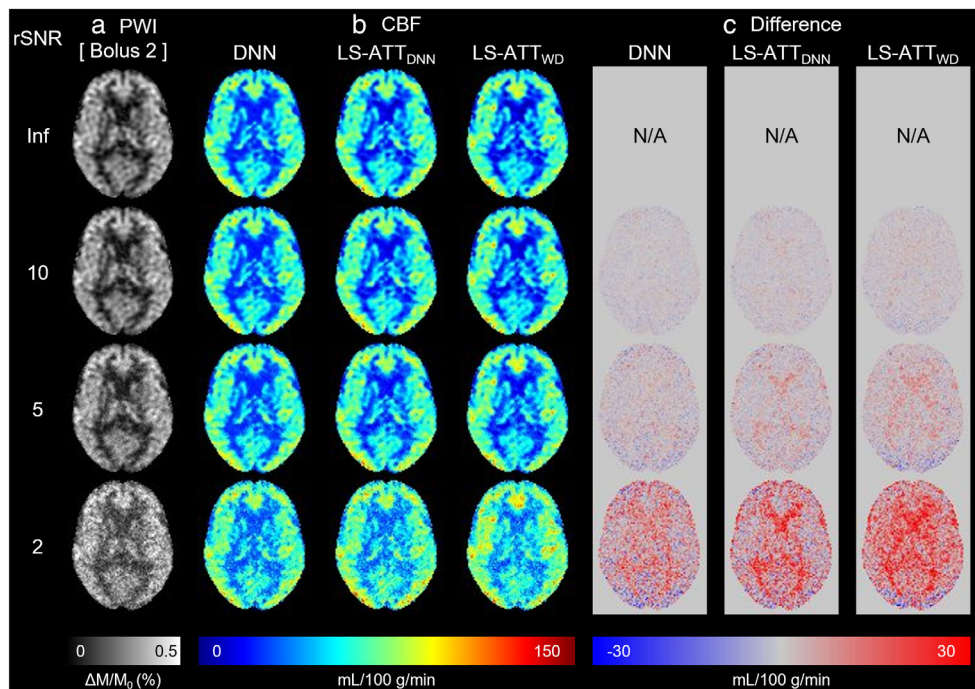


FIGURE 5: Perfusion-weighted images (PWI, a), CBF maps (b), and difference CBF maps (c) of a 58-year-old male, calculated from the noisy images synthesized at four levels of relative signal-to-noise ratio (rSNR). Infinite (Inf) rSNR means that noise is not artificially added. Bolus 2 of the combined approach (2033 msec PLD, 1333 msec LD) is representatively shown due to the closest conditions to the reference condition (2000 msec PLD, 1800 msec LD). Difference CBF maps between the Inf and respective rSNR images are also presented, where blue and red indicate under- and overestimation induced by noise. DNN, deep neural network; LS-ATT_{DNN}, least-squares using ATT estimated by DNN; LS-ATT_{WD}, least-squares using ATT estimated by signal weighted-delay method (WD); N/A, not available.

feasibility because it is a challenging pathology for perfusion quantification by ASL.³³ The combined approach was performed under the same conditions as the volunteer study, except for a higher spatial resolution (seven interleaves). With the same configuration, the selected DNNs (CBF_{opt} and ATT_{opt}) were retrained using wider ranges of the ground truth, assuming abnormal conditions (20–120 mL/100 g/min CBF, and 200–4000 msec ATT).

To compare the CBF and ATT maps between the DNN and conventional methods, difference maps were computed. Moreover, Bland–Altman graphical analysis was performed using five representative slices to compare the methods.

Statistical Analysis

GraphPad Prism 9.3 (GraphPad Software, Inc., La Jolla, CA) was used for statistical analysis. Baseline CBF and ATT were compared using a one-way analysis of variance (ANOVA) with a post-hoc Tukey's multiple comparisons test and paired *t*-test, respectively. The one-way ANOVA with post-hoc Tukey's multiple comparison test was also performed to compare NRMSE for assessing residuum between the different pairs of CBF and ATT. Statistical significance was defined as $P < 0.05$.

Moreover, Bland–Altman graphical analyses were performed to compare the predicted values and ground truth in the simulation and volunteer studies.

Results

Selection of the Best-Performing Network

Figure 3 shows the relation between each metric and the rSNR. The performance of CBF_{opt} and ATT_{opt} was superior to that of

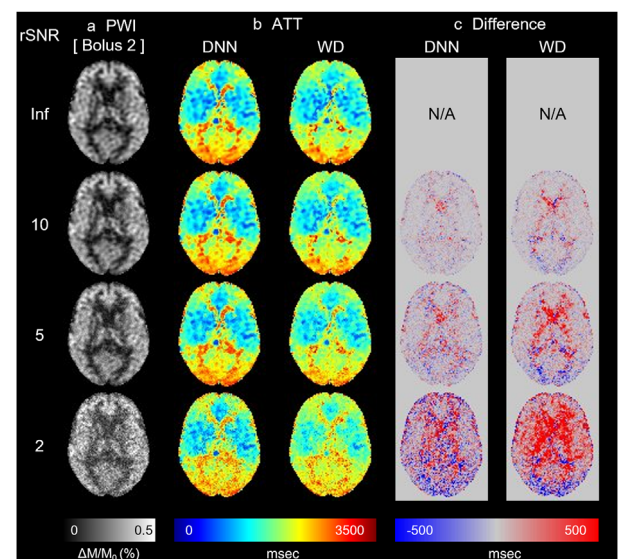


FIGURE 6: Perfusion-weighted images (PWI, a) and ATT maps (b) obtained from the same subject as Fig. 5 at four levels of relative signal-to-noise ratio (rSNR) are presented. Bolus 2 of the combined approach (2033 msec PLD, 1333 msec LD) is extracted as representative PWIs because its condition is close to the reference condition (2000 msec PLD, 1800 msec LD). Difference ATT maps (c) indicating noise effects on parameter prediction are also provided. The blue and red indicate shortened and lengthened ATT induced by artificially added noise, respectively. Infinite (Inf) rSNR means that noise is not added to the ASL images. DNN, deep neural network; WD, signal weighted-delay method; N/A, not available.

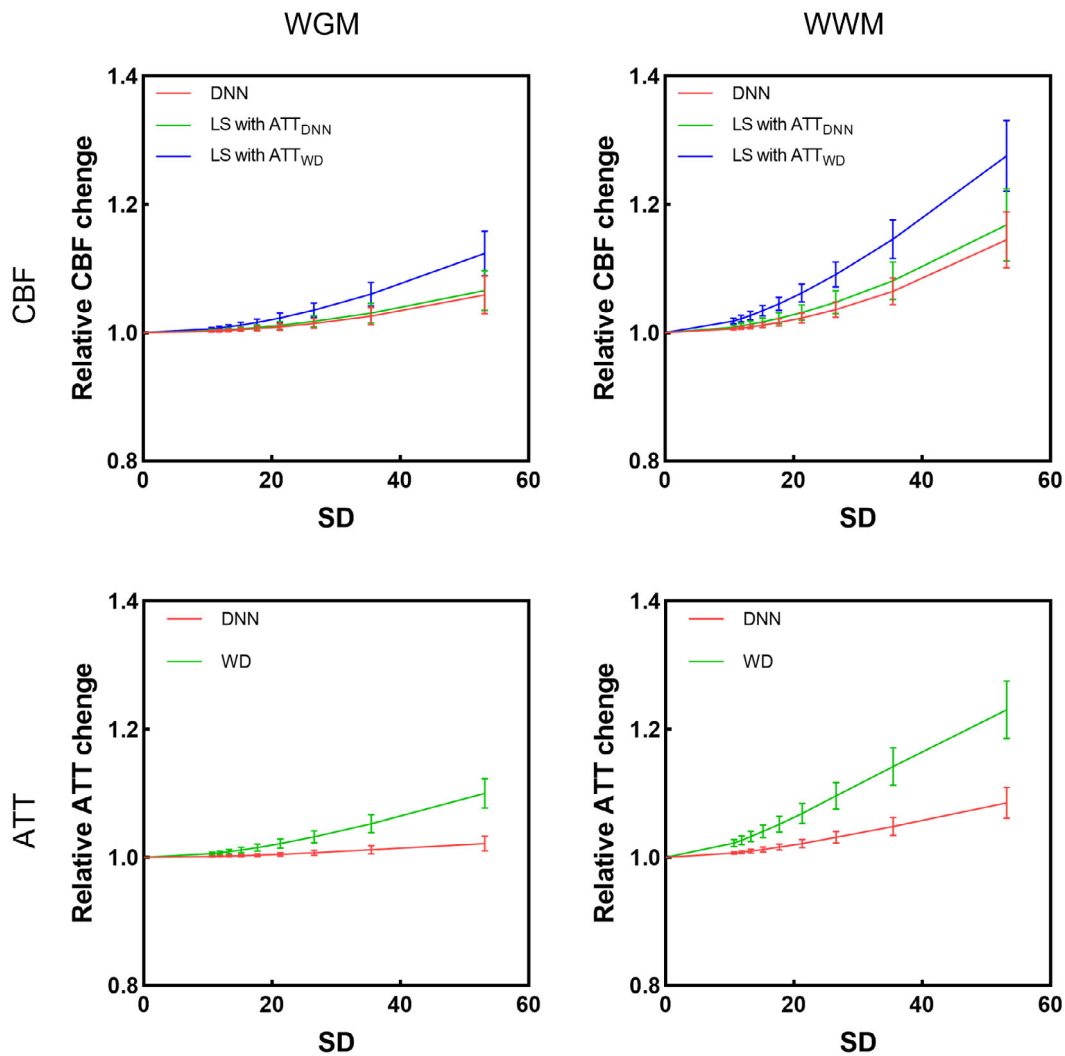


FIGURE 7: Relative changes in cerebral blood flow (CBF) and arterial transit time (ATT) against the standard deviation (SD) of artificially added noise determined in healthy volunteers. The left and right columns are the results determined in whole gray matter (WGM) and whole white matter (WWM), respectively. The upper and lower rows present CBF and ATT, respectively. A larger SD indicates a lower relative signal-to-noise ratio (rSNR). CBF and ATT without added noise (i.e., zero SD) are considered as references for assessing relative changes in CBF and ATT.

the others because of the lower CV_{Net} , NMAE, and NRMSE. Based on the summed ranking of each metric, CBF_{opt} and ATT_{opt} were selected as the best-performing networks for CBF and ATT predictions, respectively. However, all DNNs became less precise and inaccurate under low rSNR conditions because high CV_{Net} , NMAE, and NRMSE values were observed.

Comparison of the DNN and the Conventional Method: A Simulation Study

Figure 4 shows the NMAE and NRMSE of the DNN and conventional method. The DNN outperformed the conventional method owing to its lower NMAE and NRMSE values. In particular, the ATT_{DNN} had substantially lower NMAE and NRMSE than the ATT_{WD} .

The Bland–Altman analyses between the predicted values and ground truth are shown in Figs. S2 and S3 in the Supplemental Material. CBF_{DNN} showed a smaller fixed bias than the

others, particularly under low rSNR conditions (Fig. S2 in the Supplemental Material). Moreover, it had narrower limits of agreement. Conversely, the proportional bias was small for all CBFs, even under low rSNR conditions. Figure S3 in the Supplemental Material also shows that the fixed and proportional biases of ATT_{DNN} were smaller than those of ATT_{WD} . ATT_{WD} had a large proportional bias even under higher rSNR conditions, which was further pronounced under low rSNR conditions. WD underestimated and overestimated long and short ATTs, respectively. The fixed bias of ATT_{DNN} was almost zero for all the rSNR conditions; and similar to CBF, ATT_{DNN} had narrower limits of agreement.

Comparison of the DNN and the Conventional Method: A Healthy Volunteer Study

This study included 17 healthy volunteers (five women and 12 men, 33.2 ± 14.6 years). Figures 5 and 6 show representative

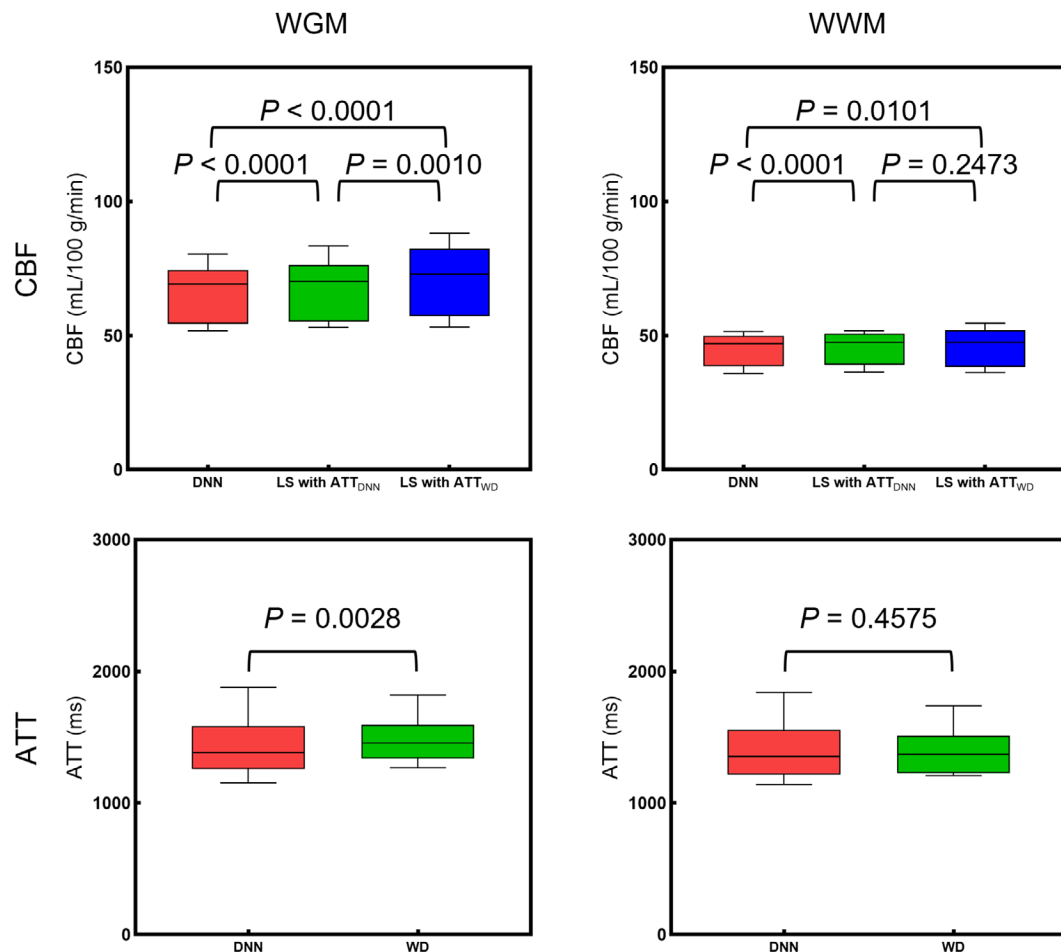


FIGURE 8: Baseline cerebral blood flow (CBF) and arterial transit time (ATT) obtained from intact images. The left and right columns show the results determined in the whole gray matter (WGM) and whole white matter (WWM), respectively. The upper and lower rows present CBF and ATT, respectively. *P*-values are provided in the box plots.

CBF and ATT maps obtained from a 58-year-old male volunteer. The synthesized noisy perfusion-weighted images (PWIs) are also presented in both figures, with intact (i.e., no artificially added noise) PWIs because of their closest conditions (2033 msec PLD; 1333 msec LD) to the reference condition that is used for rSNR definition (2000 msec PLD; 1800 msec LD). Blue and red colors in the difference maps indicate the under- and overestimated areas induced by artificial noise, respectively. For both the parameter maps, the degradation of image quality, along with a decrease in rSNR, was alleviated by the DNN. In particular, the short and long ATT regions were over- and underestimated by the WD. The improved image quality and noise immunity of the DNN were validated by quantitative assessment of the relative changes in CBF and ATT.

The relative changes in CBF and ATT with respect to the SDs are presented in Fig. 7. The relative change in the CBF_{DNN} was smaller than that of the other models. However, all CBF values increased with increasing SDs (i.e., decreasing rSNR). Similar to CBF, the relative change in ATT_{DNN} was smaller than that of ATT_{WD}. In addition, ATT_{DNN} was less sensitive to

an increase in SDs. Conversely, ATT_{WD} increased with increasing SDs.

Figure 8 shows a comparison of baseline CBF and ATT values. The CBF_{DNN} (WGM = 66 ± 10 and WWM = 45 ± 6 mL/100 g/min) was significantly smaller than those obtained by the other methods (CBF_{LS} – ATT_{DNN}, WGM = 68 ± 10 and WWM = 45 ± 6 mL/100 g/min; CBF_{LS} – ATT_{WD}, WGM = 71 ± 12 and WWM = 46 ± 7 mL/100 g/min). Similarly, in WGM, ATT_{DNN} was significantly smaller (1424 ± 201 msec) than ATT_{WD} (1471 ± 154 msec). However, the ATTs in WWM were not significantly different (ATT_{DNN} = 1394 ± 201 msec, ATT_{WD} = 1384 ± 156 msec, $P = 0.4575$).

The averaged residuum maps across all subjects after masking are shown in Fig. 9. The NRMSE in GM (NRMSE_{GM}) was smaller than that of WM (NRMSE_{WM}). As shown in the residuum maps, NRMSE_{GM} and NRMSE_{WM} using the pairs of CBF_{DNN} and ATT_{WD} were significantly larger than those obtained using the other pairs of estimators. There were no significant differences between the remaining NRMSEs.

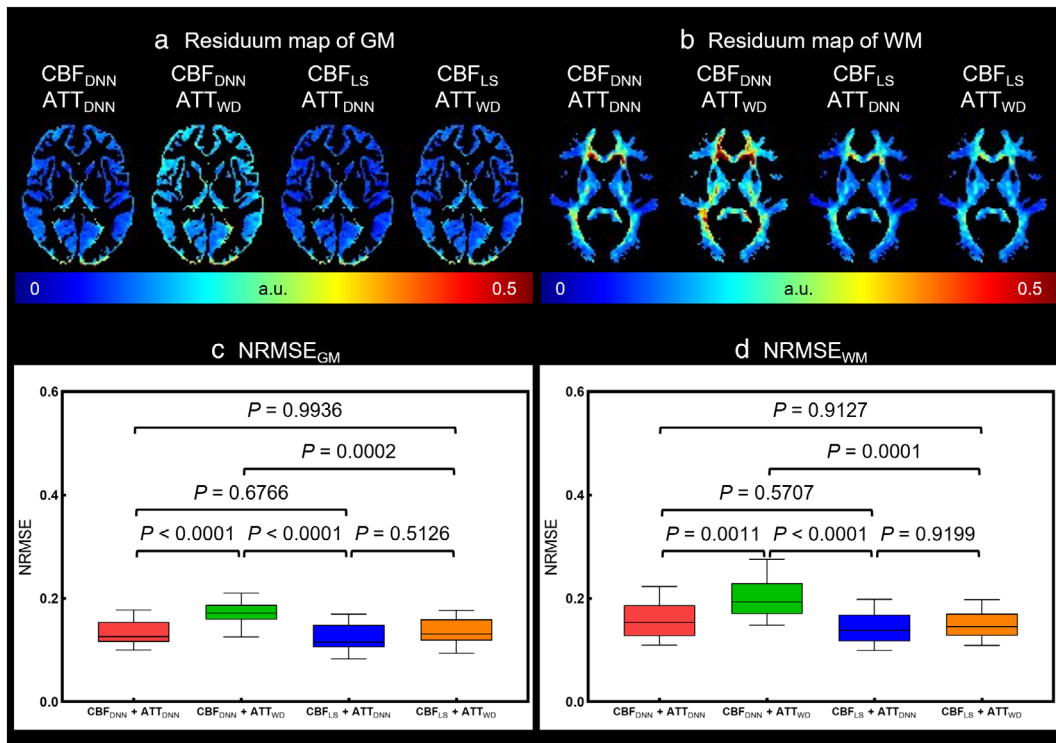


FIGURE 9: Averaged residuum maps across all subjects after gray matter (GM, a) and white matter (WM, b) masking. Box plots indicating the statistical comparison of the normalized root mean squared error (NRMSE) between the observed and the predicted ASL signals, determined in GM (c) and WM (d). *P*-values between each pair of CBF and ATT are also provided in the box plots (c) and (d), respectively.

Comparison of the DNN and the Conventional Method: A Clinical Case

Figure 10 shows a clinical case of moyamoya disease. CBF and its difference maps showed that higher CBF in the prolonged ATT region was obtained by DNN than by the conventional method. ATT and its difference maps showed that ATT_{DNN} was higher than ATT_{WD} in the prolonged regions. Moreover, ATT was shorter with DNN in the short-ATT regions. Bland–Altman graphical comparison (fourth row of Figure 10) revealed that the low and high CBF values predicted by DNN tended to be larger and smaller, respectively, than those predicted by the conventional methods. However, the low and high ATT values estimated by DNN tended to be shorter and longer, respectively, than those estimated by the conventional methods.

Discussion

In this study, neural networks were trained to improve the accuracy and precision of CBF and ATT using simulation-based supervised learning. The performance was assessed using simulation data, healthy volunteers, and a clinical patient. The best-performing network architecture and optimal configuration were investigated. The accuracy and precision of the proposed method were higher than those of the conventional method. The feasibility of DNN-based parameter estimation in a patient with moyamoya disease was also demonstrated.

Only a few studies have investigated the application of DL to ASL.^{23–27} No study focused on parameter estimation. In this respect, the current study provided new insights into the usefulness of DNNs in ASL parameter estimation.

Multiple literature values were comprehensively referenced to define the ground truth ranges.^{8,9,28} Although DNN demonstrated higher performance than that of the conventional method with the current setting, DNN performance may deteriorate for outside CBF and ATT from the ground truths. How DNN will behave for CBF and ATT outside the range of the ground truth and effects of the outside values on the optimal network configuration, accuracy, and precision should be investigated. If wider ranges of ground truth that comprise all possible CBF and ATT values are required to address this issue, effects of ground truth ranges on the performance of DNNs should also be validated.

Finer rSNR levels were set for the training dataset compared to those the test dataset to improve the prediction performance of DNNs for the test dataset and unknown data. Further studies should be performed to verify whether this approach truly improves the network performance. Effects of the number and setting of the training data on the performance of DNNs should also be investigated.

The best-performing network was selected in terms of accuracy and precision. Accuracy was evaluated by NMAE and NRMSE because it is still controversial whether MAE or

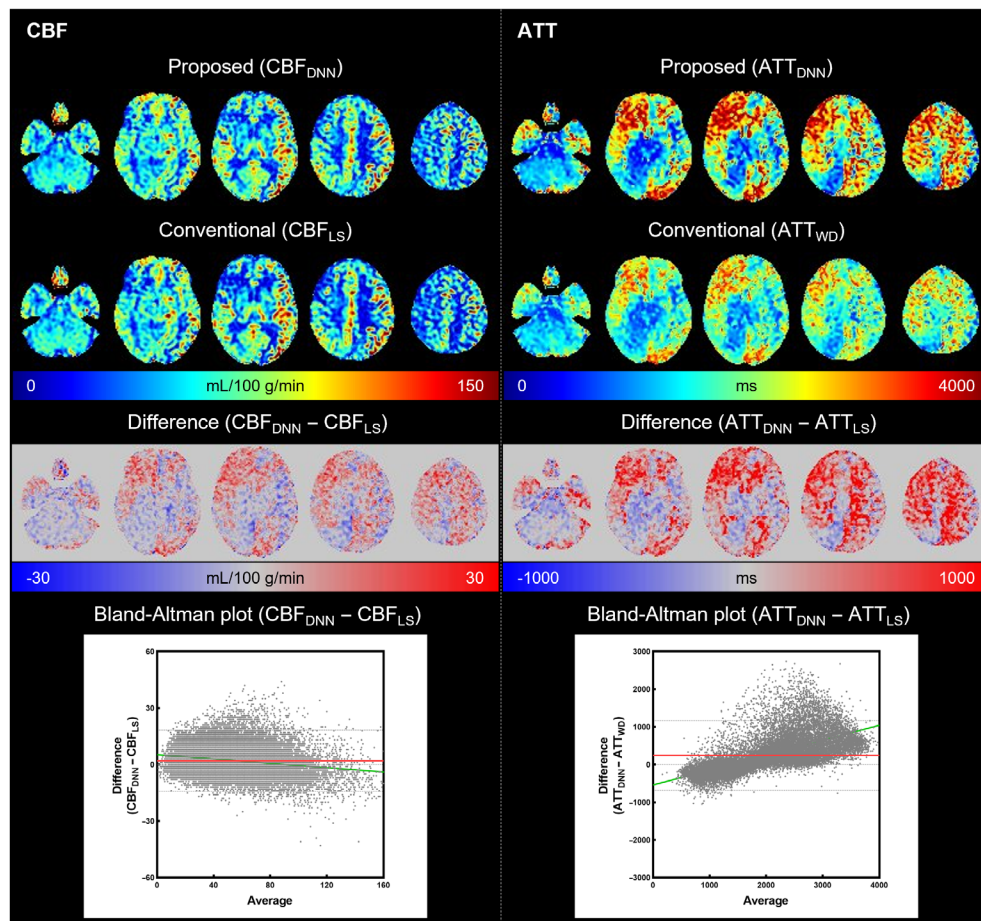


FIGURE 10: The left and right panels show CBF, and ATT maps obtained from a patient with moyamoya disease (68-year-old male), respectively. The first and second rows present the parameter maps estimated using the DNN and the conventional methods, respectively. Herein, CBF_{LS} is identical to CBF_{LS} - ATT_{WD} of the simulation and volunteer studies. The third row shows the difference parameter maps between the methods. Bland-Altman plots using five representative slices are provided at the bottom of the figure.

RMSE is better. A comprehensive assessment of multiple metrics is recommended.³⁴ Precision was assessed using a previously reported metric.²¹ The network selection procedures were considered appropriate and reasonable.

CBF_{ATT} was implemented to replicate the reconstruction process of the conventional method, in which ATT was estimated using WD before CBF calculation using the LS solution, and ATT_{CBF} followed the opposite procedure. However, their performance was inferior to those of CBF_{opt} and ATT_{opt}. Similarly, *Simultaneous* and *Parallel* networks assumed nonlinear LS fitting that concurrently estimated CBF and ATT; however, they also had lower performances than those of CBF_{opt} and ATT_{opt}. CBF_{opt} and ATT_{opt} outperformed the others, ensuring that a dedicated network for estimating a single parameter had a higher performance. Because DNN configurations are numerous, whether this assertion is completely universal is unknown. However, no similar studies have been conducted in the area of ASL parameter estimation. Therefore, based on the results of this study, the use of two separate networks optimized for single-parameter prediction is

preferable for CBF and ATT estimations from multi-delay ASL signals.

The DNN and conventional methods were compared based on NMAE and NRMSE. The CBF_{LS} - ATT_{WD} method exhibited the lowest accuracy among the three CBFs. CBF_{DNN} exhibited a slightly higher accuracy than those of CBF_{LS} - ATT_{DNN}. Furthermore, ATT_{DNN} was more accurate than ATT_{WD}. As the accuracy of ATT_{DNN} was high, that of CBF_{DNN} or CBF_{LS} - ATT_{DNN} was not significantly different because previous work showed that CBF accuracy using the LS solution strongly depended on ATT accuracy.⁹ In other words, the accuracy of CBF_{LS} - ATT_{DNN} is attributed to the accuracy of ATT_{DNN}. Thus, DNN works well for both CBF and ATT predictions over a wide range of SNR. In addition, CBF_{DNN} may be more robust than CBF_{LS} - ATT_{DNN} because it estimates CBF directly from measured signals, thereby avoiding noise propagation originating from two-step estimation of the conventional method.

Moreover, the DNN and conventional methods were compared using Bland-Altman graphical analyses. Both CBF_{DNN} and ATT_{DNN} had higher accuracy and precision

owing to the smaller fixed biases and narrower ranges of the limits of agreement. Consistent with a previous report, a lower rSNR caused larger proportional biases, particularly with ATT_{WD} .⁹ However, owing to the higher estimation performance of DNN than WD, proportional biases were smaller in ATT_{DNN} than in ATT_{WD} . Prolonged ATT and reduced SNR (i.e., hypoperfusion) are frequently encountered in clinical practice, complicating parameter estimation.³⁵ The high accuracy and noise immunity of DNNs may facilitate the clinical use of multi-delay ASL.

Each parameter map and quantitative analysis of their relative changes against the SDs revealed that CBF_{DNN} and ATT_{DNN} had high noise tolerance. Consistent with the results of the Bland–Altman plots, ATT difference maps indicated that the WD underestimated ATT in the territory of the posterior cerebral artery, whose ATT was longer than those of other regions.^{9,36} A less rough appearance of CBF and ATT maps was also achieved using DNN. This may be serendipitous because the proposed method is not a spatial-denoising technique. Relative changes in CBF and ATT increased with increasing SDs in all methods, consistent with the increased fixed bias under low rSNR conditions depicted in the Bland–Altman plots. The smaller relative changes in CBF_{DNN} and ATT_{DNN} represent the noise immunity of DNN.

The baseline CBF and ATT values were significantly smaller with DNN than those obtained via the other methods. This could be derived from the noise inherent in the original images because CBF and ATT systematically increased with noise, as shown by the results of the relative changes. Moreover, NRMSE between the observed and calculated ASL signals from the predicted estimators were compared to investigate systematic differences depending on the method. NRMSE did not differ significantly among the combinations of (CBF_{DNN} , ATT_{DNN}), (CBF_{DNN} , ATT_{WD}), and (CBF_{LS} , ATT_{DNN}). This result indicates the absence of difference in the goodness of fit among them, i.e., no systematic bias related to the technical aspects of each method was detected. Thus, no technical systematic bias originated from the methods; however, the noise inherent in images caused differences in baseline CBF and ATT. Conversely, NRMSE was higher in WM than that in GM, which was caused by two reasons. One is the lower SNR of WM, while the other is the setting of $T_{1\rho}$ assuming GM. DNN with a tissue-adaptive model may ameliorate WM performance.²²

DNNs were applied to a patient with moyamoya disease to demonstrate the clinical feasibility. The comparison results of the relative changes and baseline values of the volunteer study suggested that the higher CBF in the prolonged ATT region and a clearer contrast in ATT identified by DNN were due to the improved performance of DNN rather than technical systematic biases. In particular, the reduced CBF with LS in the prolonged ATT region was salvaged by DNN. This is a promising result for clinical applications of DNN. Although the LS solution is sensitive to intra-arterial signals,¹⁵ high CBF, which may be caused by

intra-arterial signals, has been slightly mitigated by DNN but not been eliminated. Generating simulation data using the two-compartment model, which comprises a vascular compartment,^{3,37} may enable removal without a vascular crusher.^{38–40} Therefore, a tendency observed by the Bland–Altman plots indicated that low and high CBFs were recovered and suppressed by DNN, respectively. The network configuration was not optimized for the clinical case because the same network architecture as the volunteer study was retrained anew using wider ground truth ranges. Retraining and avoiding the optimization process may reduce the estimation performance. Nevertheless, DNN possibly improved the performance of parameter estimation in a clinical case.

Limitations

First, this study used a single-center, single-vendor, single-scanner dataset that includes a small cohort and only one clinical patient. Second, a small discrepancy between the assumed and actual reconstruction processes existed. Although magnitude subtraction was assumed to generate simulation data, actual perfusion signals were computed using complex subtraction. Noise propagation stemming from the reconstruction process differed between the simulation and real scenarios. However, this difference was not a problem because the proposed method sufficiently improved the estimation performance of CBF and ATT. Complete reproduction of the reconstruction process may further improve DNN. Third, DNN was retrained using wider ground truth ranges for moyamoya disease. To justify this process theoretically, the effects of the ground truth setting on simulation-based supervised learning should be clarified; however, this was beyond the scope of the present study. Furthermore, the clinical utility of the proposed method should be validated in patients with other diseases.

Conclusion

DNN outperformed the conventional method that used the signal weighted-delay method and the least-squares solution for estimating ATT and CBF, respectively, in terms of accuracy and precision. High noise immunity was achieved using DNN-based parameter prediction. More reliable and robust parameters may be obtained using DNN.

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References

1. Detre JA, Leigh JS, Williams DS, Koretsky AP. Perfusion imaging. *Magn Reson Med* 1992;23:37–45.
2. Alsop DC, Detre JA, Golay X, et al. Recommended implementation of arterial spin-labeled perfusion MRI for clinical applications: A consensus of the ISMRM perfusion study group and the European consortium for ASL in dementia. *Magn Reson Med* 2015;73:102–116.

3. Ishida S, Kimura H, Takei N, et al. Separating spin compartments in arterial spin labeling using delays alternating with nutation for tailored excitation (DANTE) pulse: A validation study using T2-relaxometry and application to arterial cerebral blood volume imaging. *Magn Reson Med* 2022;87:1329-1345.
4. Yoo RE, Choi SH, Cho HR, et al. Tumor blood flow from arterial spin labeling perfusion MRI: A key parameter in distinguishing high-grade gliomas from primary cerebral lymphomas, and in predicting genetic biomarkers in high-grade gliomas. *J Magn Reson Imaging* 2013;38:852-860.
5. Tsujikawa T, Kimura H, Matsuda T, et al. Arterial transit time mapping obtained by pulsed continuous 3D ASL imaging with multiple post-label delay acquisitions: Comparative study with PET-CBF in patients with chronic occlusive cerebrovascular disease. *PLoS One* 2016;11:e0156005.
6. Suo X, Lei D, Cheng L, et al. Multidelay multiparametric arterial spin labeling perfusion MRI and mild cognitive impairment in early stage Parkinson's disease. *Hum Brain Mapp* 2019;40:1317-1327.
7. Takeuchi K, Isozaki M, Higashino Y, et al. The utility of arterial transit time measurement for evaluating the hemodynamic perfusion state of patients with chronic cerebrovascular stenosis or occlusive disease: Correlative study between MR imaging and (15)O-labeled H₂O positron emission tomography. *Magn Reson Med Sci* 2022.
8. Guo J, Holdsworth SJ, Fan AP, et al. Comparing accuracy and reproducibility of sequential and Hadamard-encoded multidelay pseudocontinuous arterial spin labeling for measuring cerebral blood flow and arterial transit time in healthy subjects: A simulation and in vivo study. *J Magn Reson Imaging* 2018;47:1119-1132.
9. Ishida S, Kimura H, Isozaki M, et al. Robust arterial transit time and cerebral blood flow estimation using combined acquisition of Hadamard-encoded multi-delay and long-labeled long-delay pseudocontinuous arterial spin labeling: A simulation and in vivo study. *NMR Biomed* 2020;33:e4319.
10. Woods JG, Chappell MA, Okell TW. A general framework for optimizing arterial spin labeling MRI experiments. *Magn Reson Med* 2019;81:2474-2488.
11. Woods JG, Chappell MA, Okell TW. Designing and comparing optimized pseudo-continuous arterial spin labeling protocols for measurement of cerebral blood flow. *Neuroimage* 2020;223:117246.
12. Dai W, Robson PM, Shankaranarayanan A, Alsop DC. Reduced resolution transit delay prescan for quantitative continuous arterial spin labeling perfusion imaging. *Magn Reson Med* 2012;67:1252-1265.
13. Wang DJ, Alger JR, Qiao JX, et al. Multi-delay multi-parametric arterial spin-labeled perfusion MRI in acute ischemic stroke - comparison with dynamic susceptibility contrast enhanced perfusion imaging. *Neuroimage Clin* 2013;3:1-7.
14. Wang R, Yu S, Alger JR, et al. Multi-delay arterial spin labeling perfusion MRI in moyamoya disease--comparison with CT perfusion imaging. *Eur Radiol* 2014;24:1135-1144.
15. Amemiya S, Watanabe Y, Takei N, et al. Arterial transit time-based multidelay combination strategy improves arterial spin labeling cerebral blood flow measurement accuracy in severe steno-occlusive diseases. *J Magn Reson Imaging* 2022;55:178-187.
16. Dai W, Shankaranarayanan A, Alsop DC. Volumetric measurement of perfusion and arterial transit delay using hadamard encoded continuous arterial spin labeling. *Magn Reson Med* 2013;69:1014-1022.
17. Positano V, Meloni A, Santarelli MF, et al. Deep learning staging of liver iron content from multiecho MR images. *J Magn Reson Imaging* 2022.
18. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. *arXiv* 2014;1409.1556.
19. Bertleff M, Domsch S, Weingartner S, et al. Diffusion parameter mapping with the combined intravoxel incoherent motion and kurtosis model using artificial neural networks at 3 T. *NMR Biomed* 2017;30:e3833.
20. Barbieri S, Gurney-Champion OJ, Klaassen R, Thoeny HC. Deep learning how to fit an intravoxel incoherent motion model to diffusion-weighted MRI. *Magn Reson Med* 2020;83:312-321.
21. Kaandorp MPT, Barbieri S, Klaassen R, et al. Improved unsupervised physics-informed deep learning for intravoxel incoherent motion modeling and evaluation in pancreatic cancer patients. *Magn Reson Med* 2021;86:2250-2265.
22. Zhou XX, Wang XY, Liu EH, et al. An unsupervised deep learning approach for dynamic-exponential Intravoxel incoherent motion MRI modeling and parameter estimation in the liver. *J Magn Reson Imaging* 2022;56(3):848-859.
23. Gong E, Guo J, Liu J, et al. Deep learning and multi-contrast based denoising for low-SNR arterial spin labeling (ASL) MRI. In *Proceedings of SPIE, Medical Imaging*. 2020; 11313.
24. Xie D, Li Y, Yang H, et al. Denoising arterial spin labeling perfusion MRI with deep machine learning. *Magn Reson Imaging* 2020;68:95-105.
25. Guo J, Gong E, Fan AP, Goubran M, Khalighi MM, Zaharchuk G. Predicting (15)O-Water PET cerebral blood flow maps from multi-contrast MRI using a deep convolutional neural network with evaluation of training cohort bias. *J Cereb Blood Flow Metab* 2020;40:2240-2253.
26. Shyna A, Ushadevi Amma C, Ansamma J, Kesavadas C, Thomas B. Deep-ASL enhancement technique in arterial spin labeling MRI - A novel approach for the error reduction of partial volume correction technique with linear regression algorithm. *J Comp Sci* 2022;58:101546.
27. Luciw NJ, Shirzadi Z, Black SE, Goubran M, MacIntosh BJ. Automated generation of cerebral blood flow and arterial transit time maps from multiple delay arterial spin-labeled MRI. *Magn Reson Med* 2022;88:406-417.
28. Fantini S, Sassaroli A, Tgavalekos KT, Kornbluth J. Cerebral blood flow and autoregulation: Current measurement techniques and prospects for noninvasive optical methods. *Neurophotonics* 2016;3:031411.
29. Clevert DA, Unterthiner T, Hochreiter S. Fast and accurate deep network learning by exponential linear units (ELUs). *arXiv* 2016;1511.07289.
30. Kingma DP, Ba J. Adam: A method for stochastic optimization. *arXiv* 2017;1412.6980.
31. He K, Zhang X, Ren S, Sun J. Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification. *arXiv* 2015; 1502.01852.
32. Dai W, Garcia D, de Bazelaire C, Alsop DC. Continuous flow-driven inversion for arterial spin labeling using pulsed radio frequency and gradient fields. *Magn Reson Med* 2008;60:1488-1497.
33. Donahue MJ, Achten E, Cogswell PM, et al. Consensus statement on current and emerging methods for the diagnosis and evaluation of cerebrovascular disease. *J Cereb Blood Flow Metab* 2018;38:1391-1417.
34. Chai T, Draxler RR. Root mean square error (RMSE) or mean absolute error (MAE)? - Arguments against avoiding RMSE in the literature. *Geosci Model Dev* 2014;7:1247-1250.
35. MacIntosh BJ, Lindsay AC, Kyliantiras I, et al. Multiple inflow pulsed arterial spin-labeling reveals delays in the arterial arrival time in minor stroke and transient ischemic attack. *AJNR Am J Neuroradiol* 2010;31:1892-1894.
36. Fujiwara Y, Matsuda T, Kanamoto M, et al. Comparison of long-labeled pseudo-continuous arterial spin labeling (ASL) features between young and elderly adults: Special reference to parameter selection. *Acta Radiol* 2017;58:84-90.
37. Wang J, Alsop DC, Li L, et al. Comparison of quantitative perfusion imaging using arterial spin labeling at 1.5 and 4.0 tesla. *Magn Reson Med* 2002;48:242-254.
38. Matsuda T, Kimura H, Kabasawa H, Kanamoto M. Three-dimensional arterial spin labeling imaging with a DANTE preparation pulse. *Magn Reson Imaging* 2018;49:131-137.
39. Fujiwara Y, Kimura H, Ishida S, et al. Intravascular signal suppression and microvascular signal mapping using delays alternating with nutation for tailored excitation (DANTE) pulse for arterial spin labeling perfusion imaging. *Magma* 2020;33:367-376.
40. Chappell MA, MacIntosh BJ, Donahue MJ, Gunther M, Jezard P, Woolrich MW. Separation of macrovascular signal in multi-inversion time arterial spin labelling MRI. *Magn Reson Med* 2010;63:1357-1365.